

Andrew Xiao

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EDUCATION

Rutgers University

New Brunswick, NJ

Bachelor of Arts in Computer Science, Minor in Economics

Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Discrete Math I & II, Linear Algebra, Data Management, Design and Analysis of Computer Algorithms

EXPERIENCE

Software Engineering Intern

Jun 2025 – Aug 2025

Mad Realities

Manhattan, NY

- Developed a RAG-based SMS chatbot integrating Community webhooks, Cloudflare, and Gemini to **3,000+** users and increasing fan interaction by **40%**
- Built an analytics dashboard in TypeScript, SQL, and Palantir Foundry, replacing **90%** of manual reporting with automated data pipelines.
- Cut Palantir Foundry dashboard loading time from **5s to 1s** by parallelizing a synchronous request waterfall.
- Led development of *The Love Channel* website on Next.js hosted on Cloudflare Workers reaching 2400+ visits.

PROJECTS

RU My Valentine | Python, pgvector, FastAPI, Pydantic, PostgreSQL, AWS EC2

Oct 2025 – Feb 2026

- Released a Valentine's Day matching platform to **2,000+** Rutgers students, over a 7 day sign up period
- Built the matching engine on pgvector L2 similarity search with HNSW indexing, encoding questionnaire compatibility responses into 70 dimensional preference vectors.
- Built an asynchronous FastAPI backend with Pydantic validation, layering JWT, OAuth, and rate-limiting.
- Enforced bidirectional matching with a 5 match cap per user and 3 match minimum.

ShuttleRank | Python, React Native, PostgreSQL, Redis, Neon, AWS EC2, Caddy

Mar 2026 – Present

- Created a Glicko-2 rating engine using a custom score differential factor with tanh saturation, tracking player uncertainty and volatility across matches.
- Validated with 149 pytest cases and a 5,000 match simulation across 200 players, revealing Pearson $r \approx 0.95$ correlation
- Added a Redis leaderboard cache, cutting repeat read times from approximately **50ms to 3ms** in Postgres
- Built a cross-platform badminton rating app on React Native over a shared PostgreSQL and Redis backend.

PyLog | Python, asyncio, Protocol Buffers, Docker, Prometheus

Aug 2026 - Present

- Built an append only log broker with segment based storage and sparse offset indexing for fast seeks.
- Added consumer groups with durable offset commits and at least once delivery across partitions.
- Designed a length prefixed binary wire protocol over asyncio TCP for concurrent producers.
- Recovered from crashes using CRC validation, torn tail truncation, and offset index rebuilds.

Menu Translator | Python, Streamlit, Tesseract OCR, OpenAI API

Aug 2026

- Built an OCR to LLM pipeline extracting menu text via Tesseract, generating translations and dish explanations.
- Engineered an image preprocessing stage to improve OCR reliability on angled, dim photos of real menus.
- Shipped a live camera capture Streamlit app, tested against 20+ menus across 5 languages with **100%** accuracy.

LEADERSHIP

Club Manager

Oct 2022 – Jul 2024

New Jersey Badminton Club

Montville, NJ

- Implemented a membership system that increased revenue flow by 20% and boosted membership growth by 15%.
- Organized a 200+ player tournament, ensuring smooth court allocation and efficient time management.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, HTML, CSS

Frameworks: FastAPI, React, Next.js, React Native, gRPC, SQLAlchemy, Pydantic, Streamlit

Infrastructure: Docker, AWS EC2, Google Cloud, PostgreSQL, pgvector, Redis, Neon, Caddy, Cloudflare Workers

Tools: Git, GitHub Actions, Linux, pytest, REST APIs, CI/CD, Tesseract OCR, OpenAI API

Activities: HackRU, Division 1 Collegiate Badminton Team